

Name: _____

Date: _____

RATIO, (BEST VALUE PROBLEMS)

LO: I can compare unit prices to determine which product offers the best value.

Best value problems

Best value means finding which option gives you the most for your money. Compare by finding the price per unit (or amount per penny).

Method: Divide price by quantity to find cost per unit. The lowest unit price is the best value.

Quick examples

●	500ml for £2 = 0.4p/ml	price per ml
●	750ml for £2.70 = 0.36p/ml	cheaper per ml
●	Lower unit price = better value	comparison rule
●	The cheapest item is always best value	must compare unit price
●	Bigger is always better value	not necessarily

Key idea

Calculate the cost per unit for each option. The one with the lowest unit cost is the best value.

$$\text{Unit price} = \frac{\text{cost}}{\text{quantity}}$$

Compare unit prices: lowest = best value

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – two sizes

Juice: 500ml for £1.20 or 1.5L for £3.00. Which is better value?

Small: $120\text{p} \div 500 = 0.24\text{p per ml}$
 Large: $300\text{p} \div 1500 = 0.20\text{p per ml}$
 Large is cheaper per ml

1.5L is better value

Compare price per ml to find the best deal.

Example 2 – three options

Cereal: 400g for £2.80, 750g for £4.50, 1kg for £6.50. Best value?

400g: $280 \div 400 = 0.70\text{p/g}$
 750g: $450 \div 750 = 0.60\text{p/g}$
 1kg: $650 \div 1000 = 0.65\text{p/g}$

750g is best value (0.60p/g)

The biggest size is not always the best value.

Example 3 – with special offer

Milk: 2 pints for £1.10 or 4 pints for £1.80 (buy 2 get 10% off). Which?

2 pints: $110 \div 2 = 55\text{p per pint}$
 4 pints with 10% off: £1.62, so $162 \div 4 = 40.5\text{p per pint}$
 4 pints with offer is better value

4 pints (with offer) is better value

Apply the discount before calculating unit price.

Example 4 – amount per pound

Shop A: 6 apples for £1.50. Shop B: 10 apples for £2.80. Best value?

Shop A: $150 \div 6 = 25\text{p per apple}$
 Shop B: $280 \div 10 = 28\text{p per apple}$
 Shop A is cheaper per apple

Shop A is better value

You can also compare apples per pound if preferred.

YOUR TURN – PRACTICE (deliberate practice)

1. Comparing two sizes

Find the unit price and state which is better value.

- a) Water: 500ml for 80p or 1.5L for £2.10
- b) Rice: 1kg for £1.80 or 2kg for £3.20
- c) Pasta: 500g for £1.35 or 1kg for £2.50
- d) Soap: 3 bars for £1.50 or 8 bars for £3.60

2. Three options

Find the best value from three options.

- a) Juice: 250ml £0.60, 500ml £1.10, 1L £1.90
- b) Crisps: 25g 50p, 50g 85p, 150g £2.10
- c) Eggs: 6 for £1.50, 10 for £2.30, 15 for £3.60
- d) Pens: 3 for £2.40, 5 for £3.75, 10 for £7.00

3. With offers and discounts

Include the offer in your calculation.

- a) A: £3.60 for 6, B: £4.80 for 8 with 25% off. Better value?
- b) A: 2L for £2.20, B: 3L for £3.60 buy-one-get-one-free. Better value?
- c) A: 500g at £4.00, B: 750g at £5.40 with 10% off. Better value?
- d) A: 12 pack £6, B: 20 pack £9 with buy 2 get 1 free. Better value?

4. Real-world best value

Solve each best value problem.

- a) Taxi A: £3 + £1.50/km. Taxi B: £5 + £1.00/km. Which is cheaper for 8 km?
- b) Phone plan A: £15/month + 5p/text. B: £20/month unlimited. Crossover texts?
- c) Petrol A: 145.9p/L. Petrol B: 142.5p/L but 3 miles further. Saving on 40L?
- d) Gym A: £30/month. Gym B: £5 per visit. Break-even visits per month?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Best value means lowest unit price ☐

Correct answer: _____

Reason: _____

b) The biggest pack is always best value ☐

Correct answer: _____

Reason: _____

c) To compare, divide price by quantity ☐

Correct answer: _____

Reason: _____

d) The cheapest total price is always the best value ☐

Correct answer: _____

Reason: _____

e) Buy-one-get-one-free halves the unit price ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A family needs **5 litres of milk** per week. Options: 1 pint (568ml) for **55p**, 2 pints for **£1.00**, 4 pints for **£1.75**, 6 pints for **£2.40**. What is the cheapest way to buy at least 5 litres?

Expression: _____

Simplified: _____

b) Two phone contracts: **Plan A**: £25/month, 10GB data, unlimited calls. **Plan B**: £18/month, 5GB data, 500 minutes. Extra data costs £5/GB, extra minutes cost 3p/min. Which plan is better value if you use **8GB data** and **400 minutes** per month?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

